



Torque Vectoring

Improving Race Vehicle Performance by
Active Yaw Moment and Traction Control

FSG Academy 2020/2021 powered by



Agenda

1. Introduction & Overview
2. Drivetrain
3. Vehicle Kinematics
4. Suspension
5. Tires
6. Sensors
7. Driving Situations
8. Vehicle Dynamics Control
9. Modeling
10. Torque Distribution
11. Traction Control
12. Tuning
13. Closing Remarks



Speakers



Erik Sulimma

2019 Cooling
2020 Vehicle Dynamics Control
2021 LiDAR SLAM



Yann Bernard

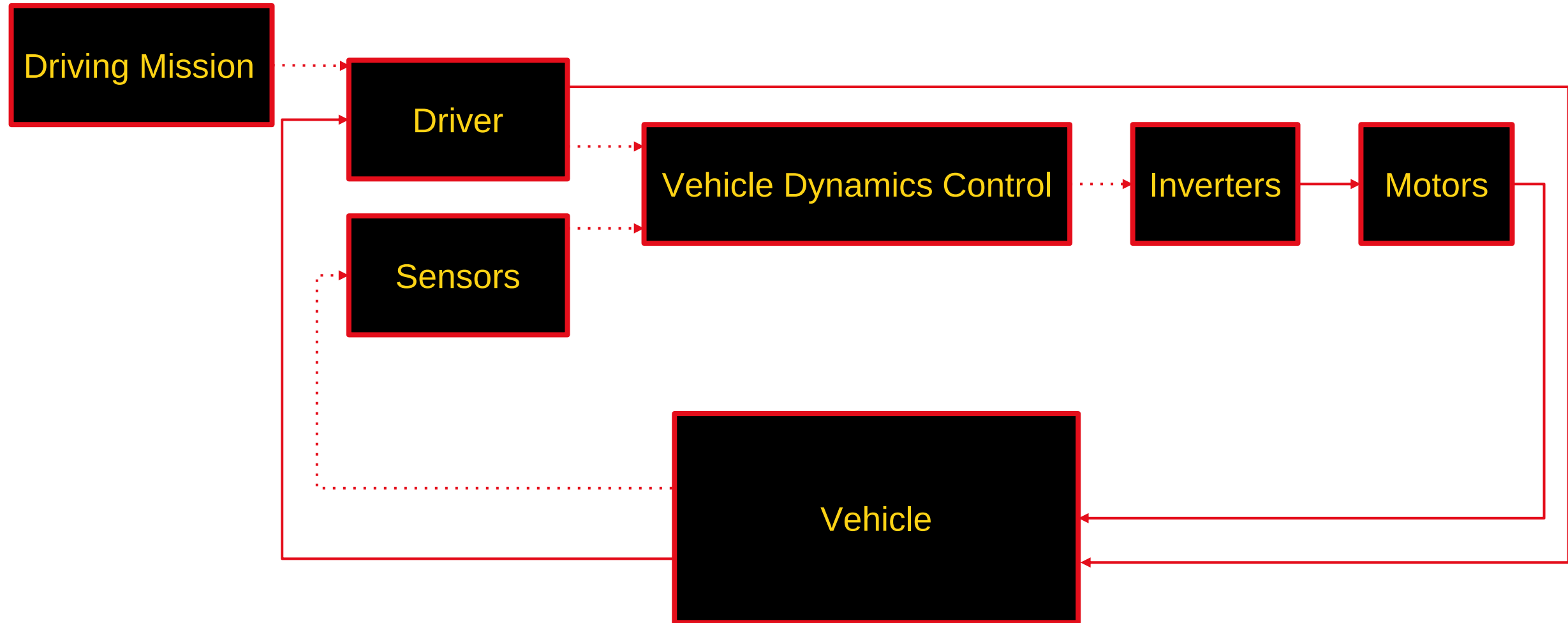
Suspension
Drivetrain & Manufacturing
CEO

Introduction: Torque Vectoring

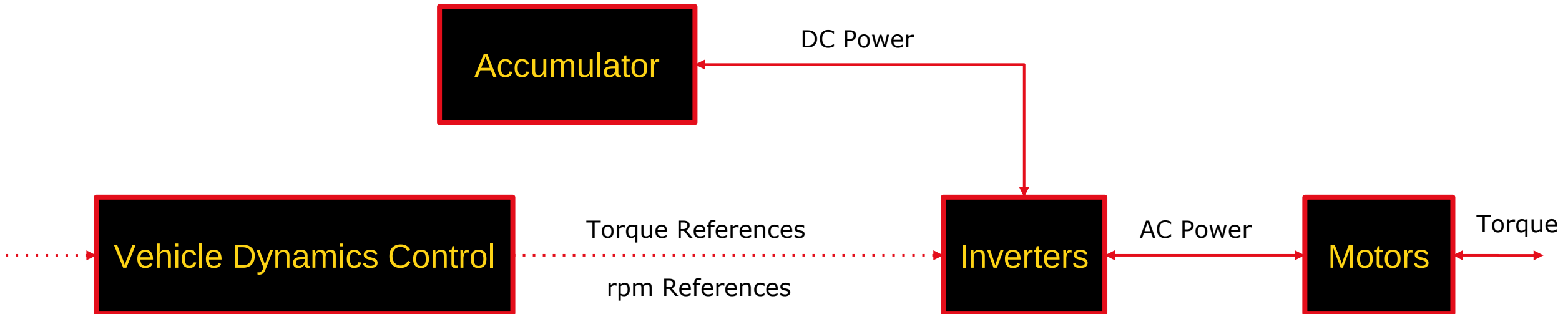
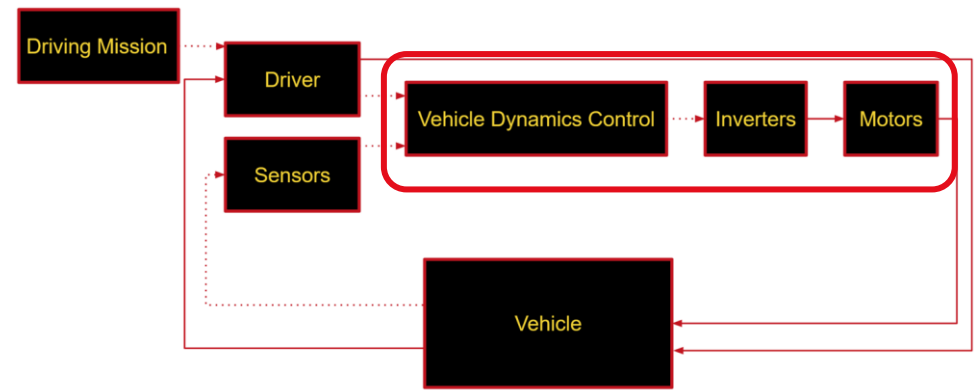
- Torque Distribution
 - Improving Vehicle Performance using active Yaw moment control
 - Driving at Edge of Tire Limits
- Traction Control
 - Driving at Optimal Slip
- Recuperation
 - Electrical Brake Balance



Overview

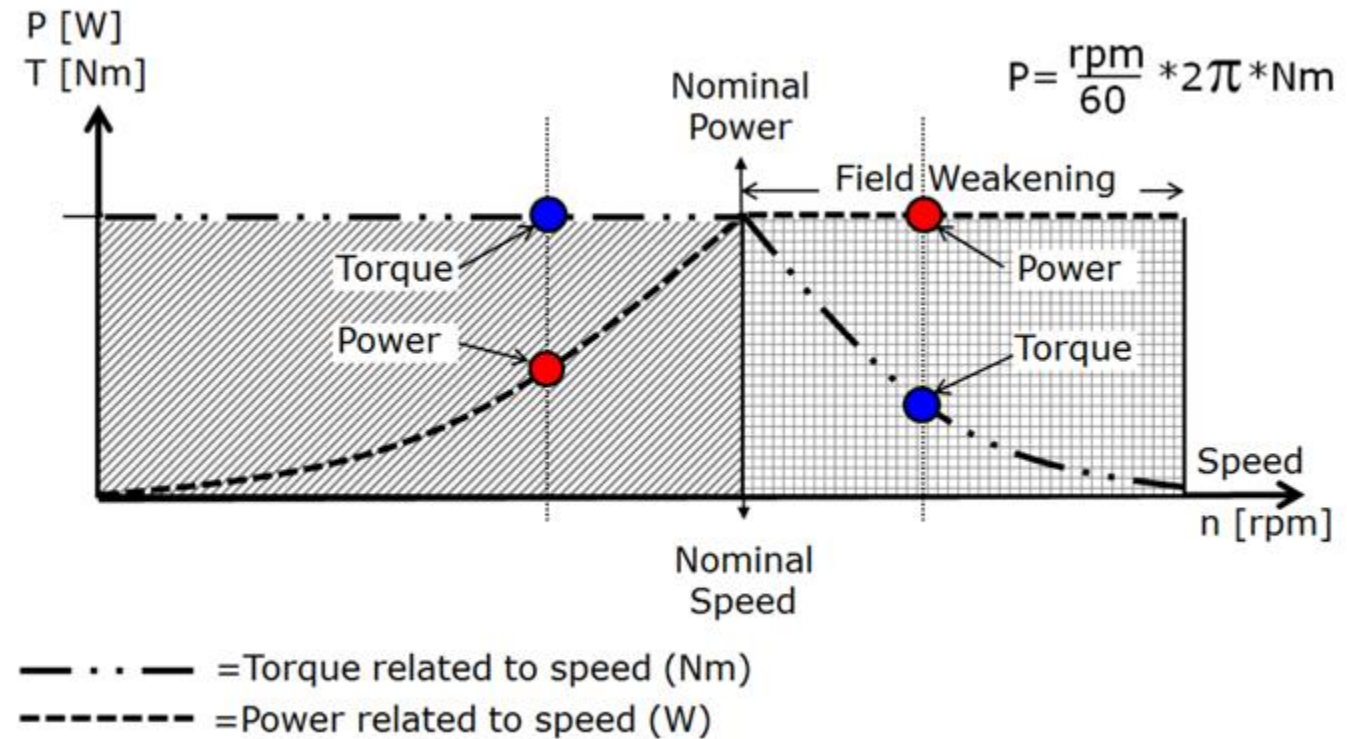


Drivetrain

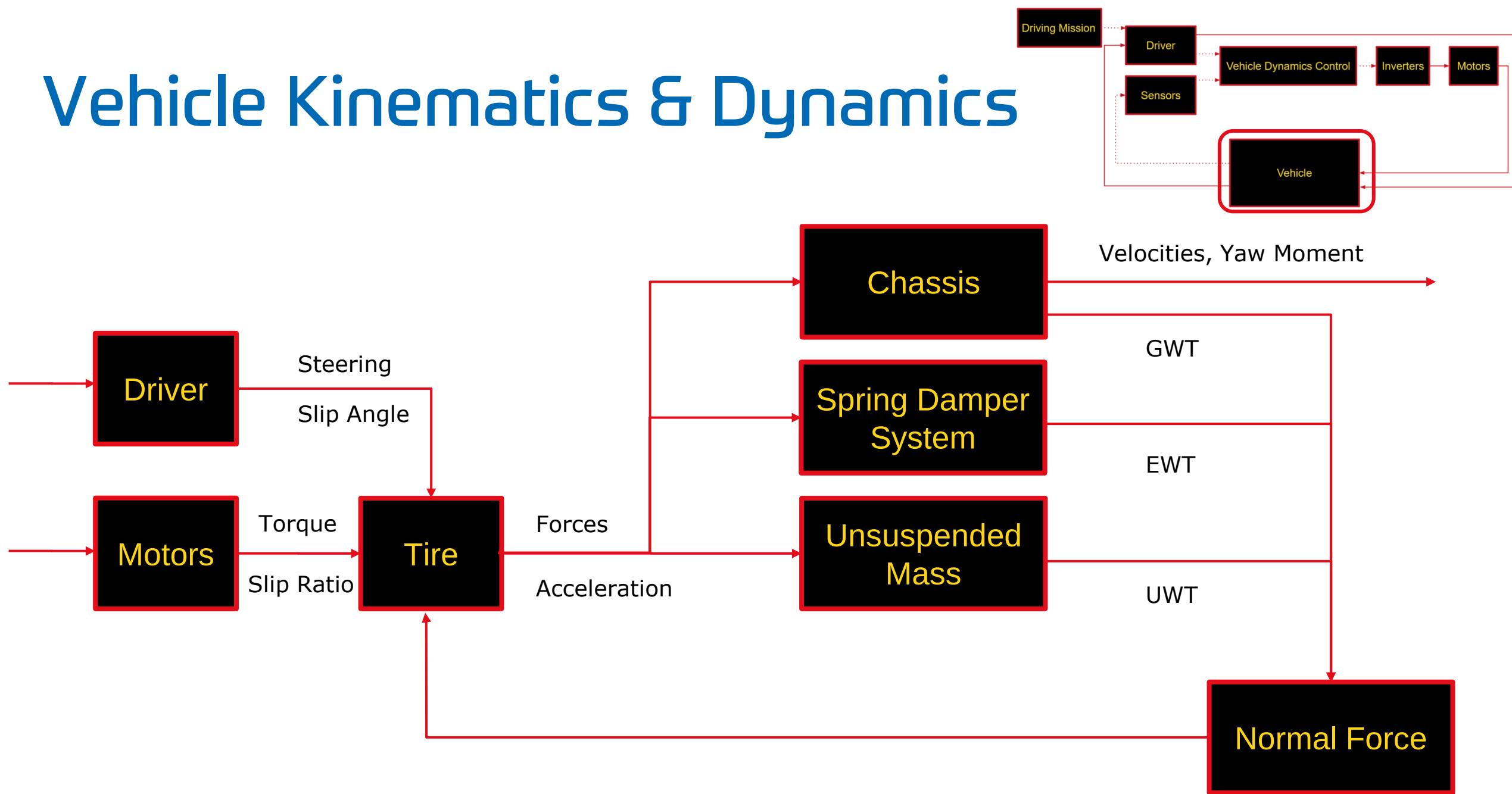


Drivetrain

- Accumulator
 - 80 kW Power Limit
 - Recuperation Power Limit
 - SoC Voltage Limit
- Inverter
 - Field Orientated Control
- Motor
 - Current Limit
 - Voltage Limit

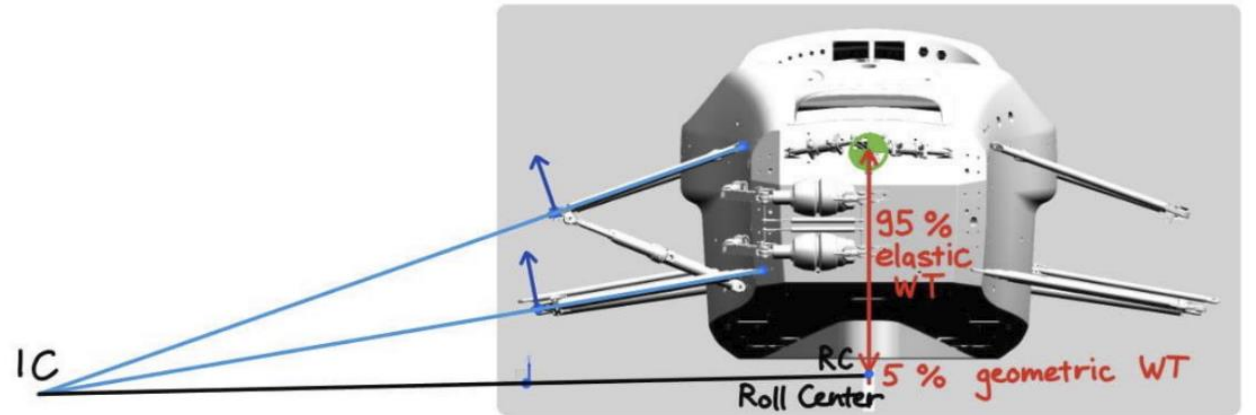
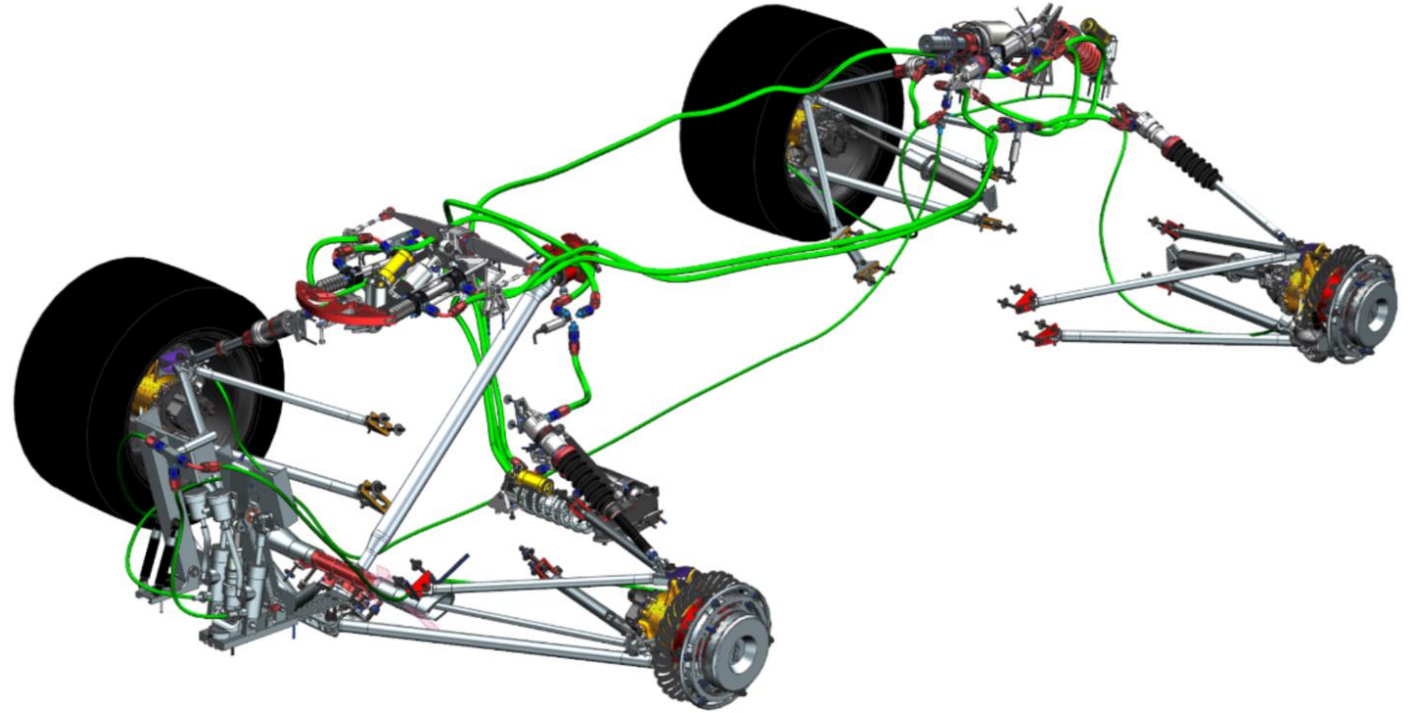


Vehicle Kinematics & Dynamics



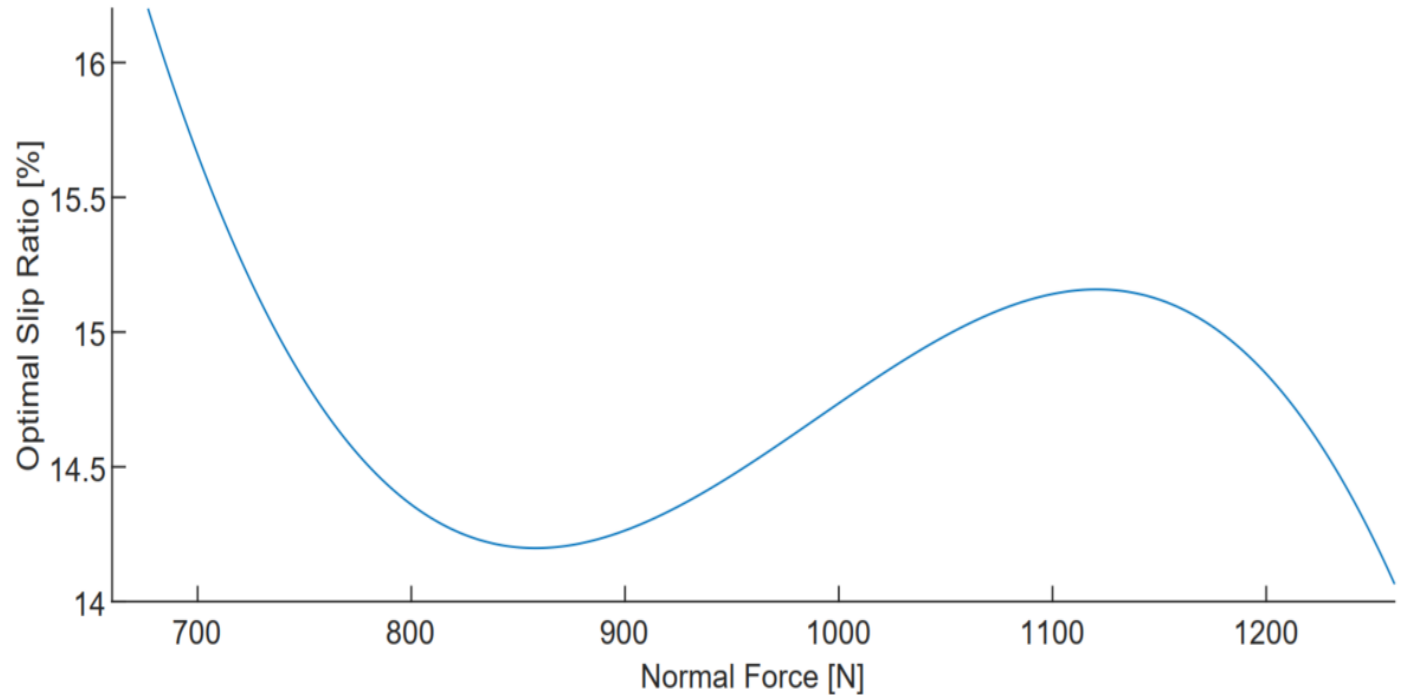
Suspension

- Pitch Center
- Roll Axis
- Weight Transfer
- Motion Ratios
- Spring Rate
- Damping Coefficients

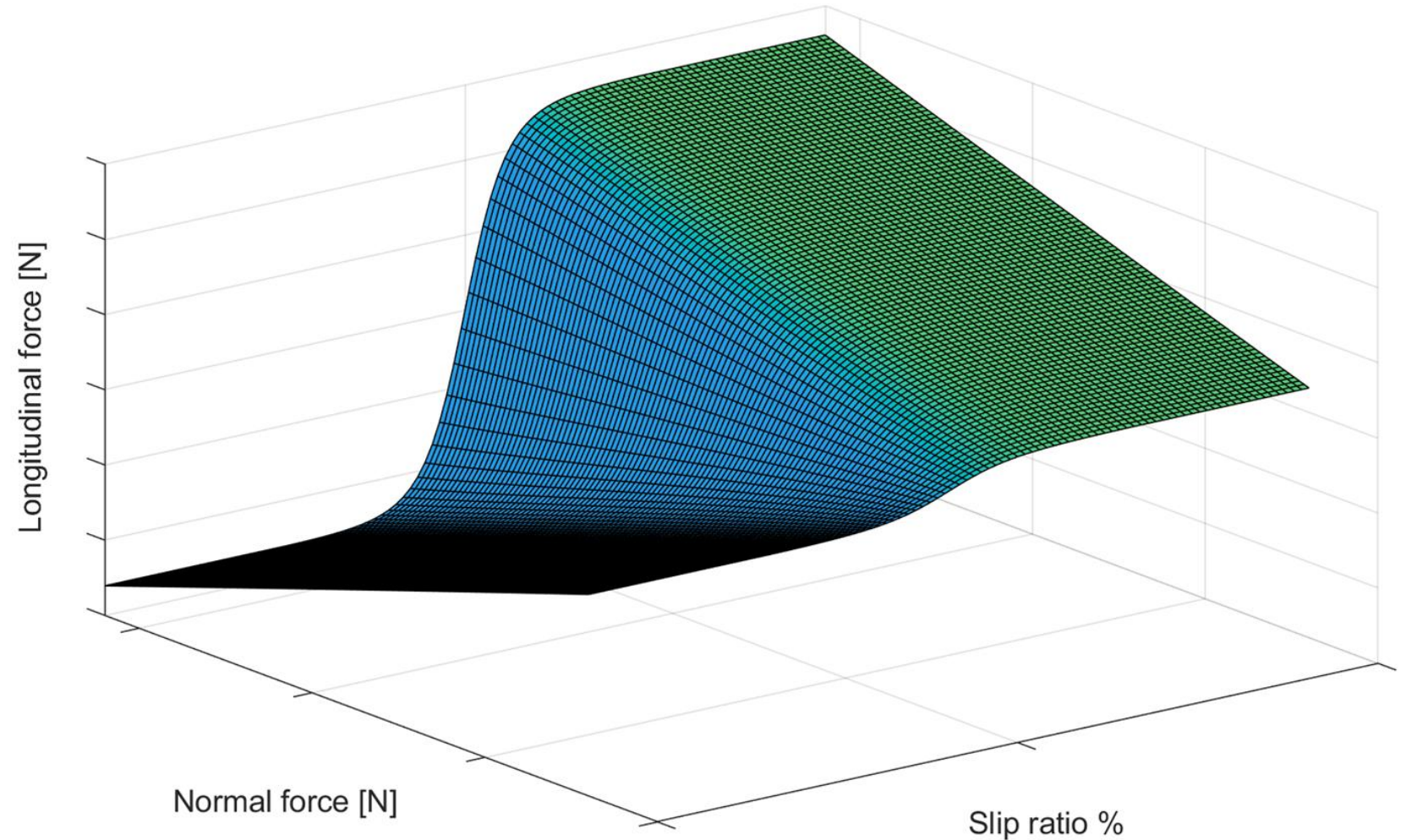


Tires

- Optimal SR
 - Normal Force
- Tire Forces
 - Normal Force and SR
 - Normal Force and SA
- Tire Limit
 - Normal Force and SR

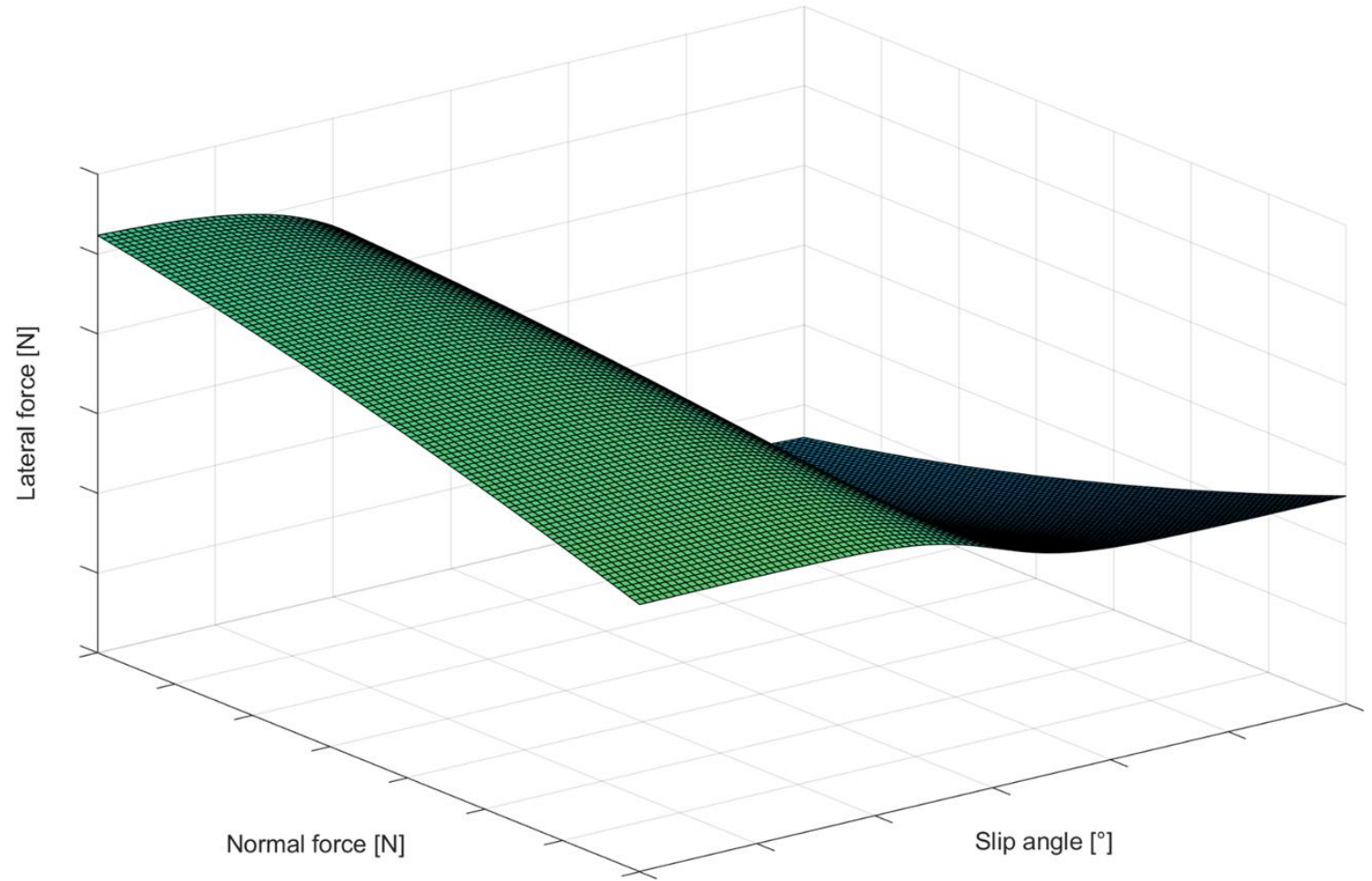
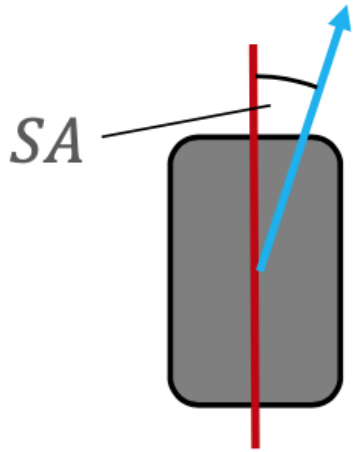


Slip Ratio



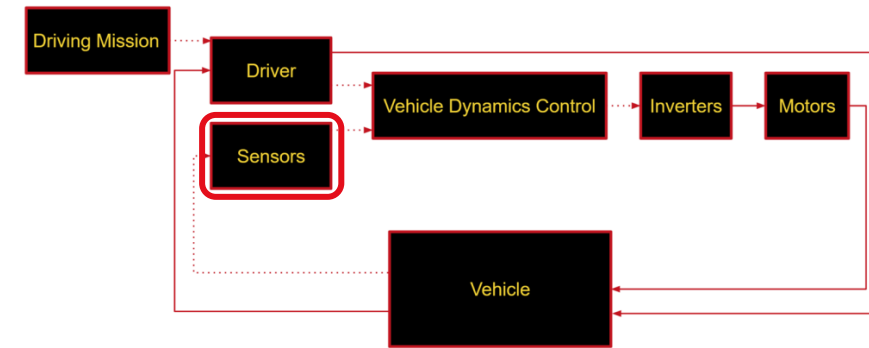
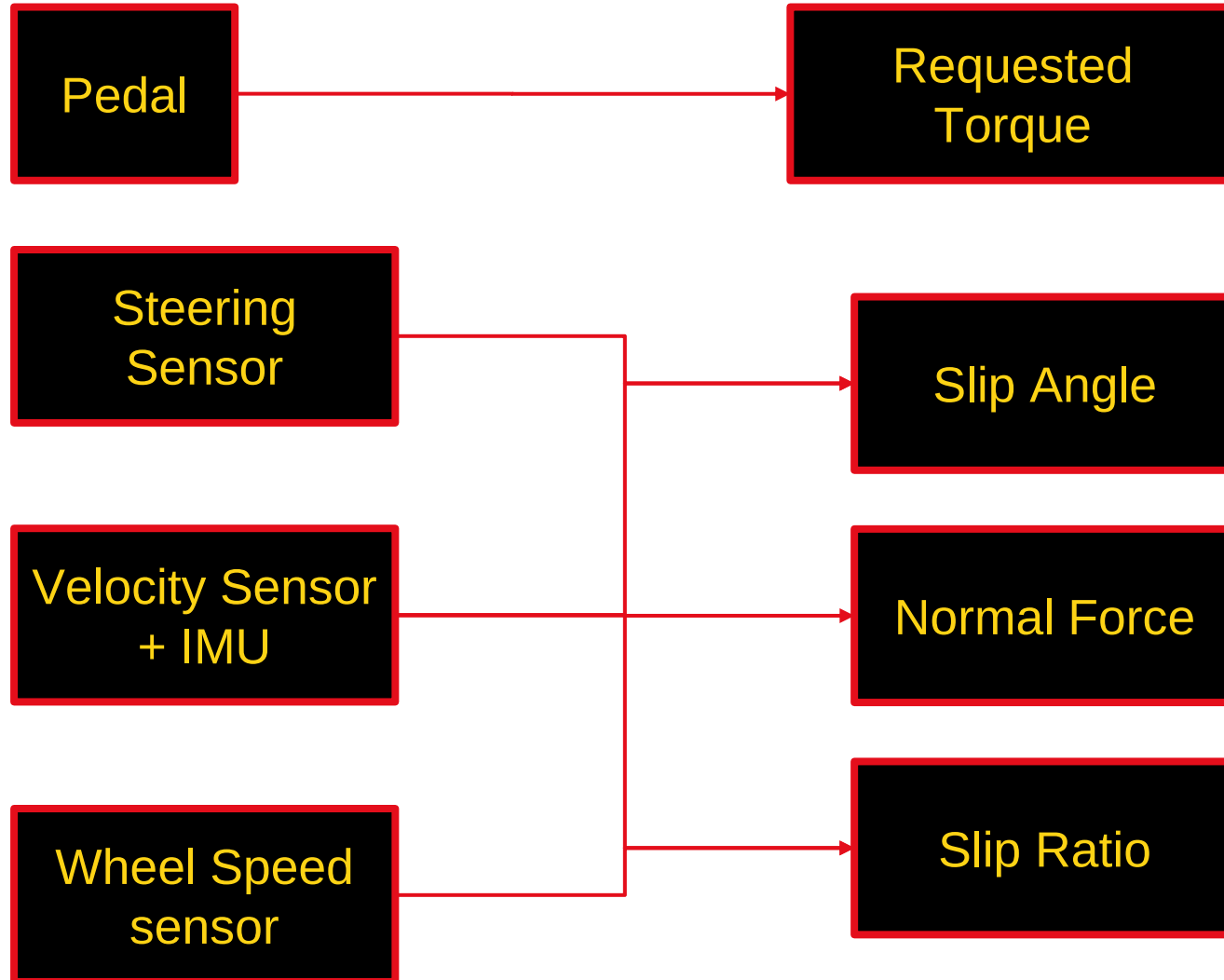
$$SR = \frac{\omega_{wheel} * r_{dyn} - v_x}{v_x}$$

Slip Angle



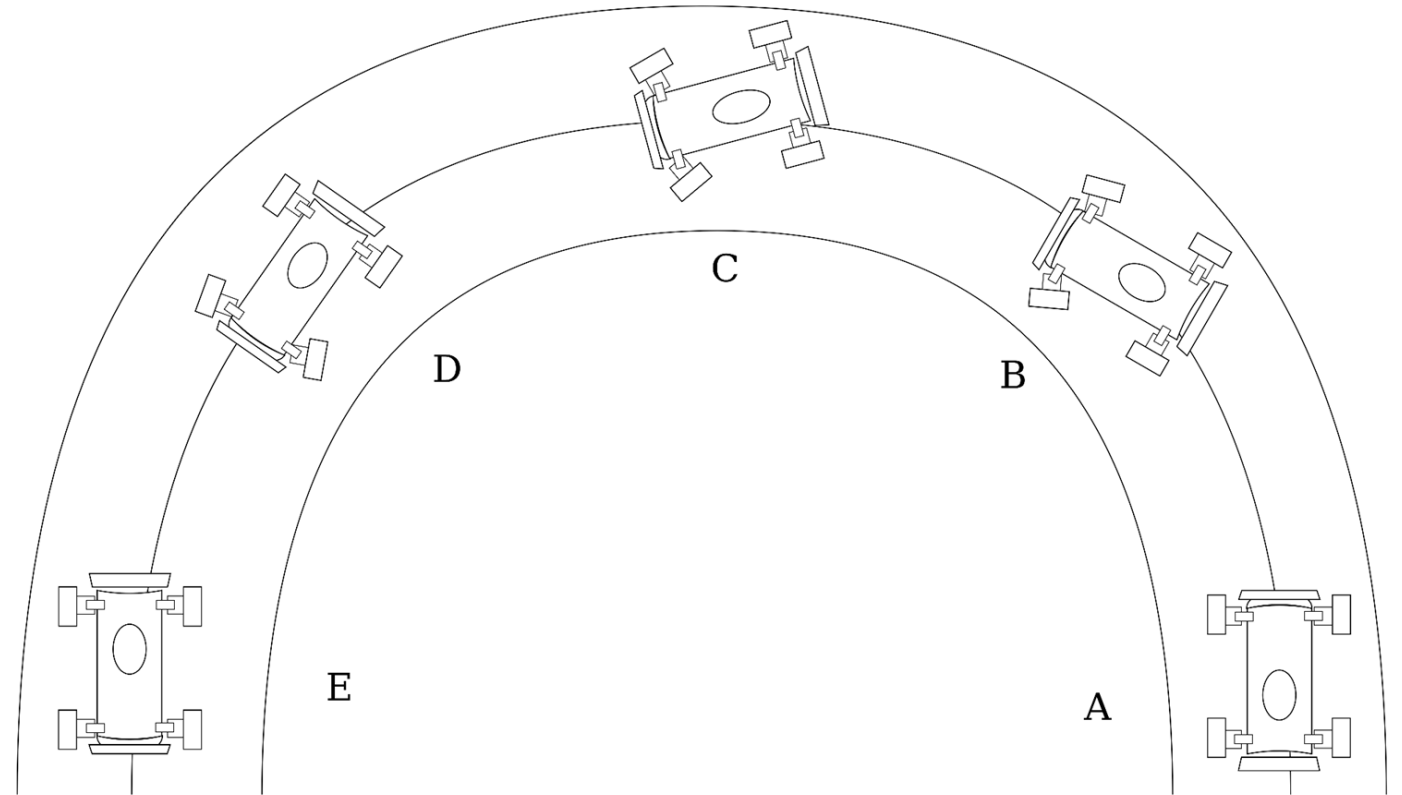
$$SA_{FL} = \frac{v_y - \omega_z \cdot l_f}{v_x - \omega_z \cdot t/2} - \delta_{FL}$$

Sensors

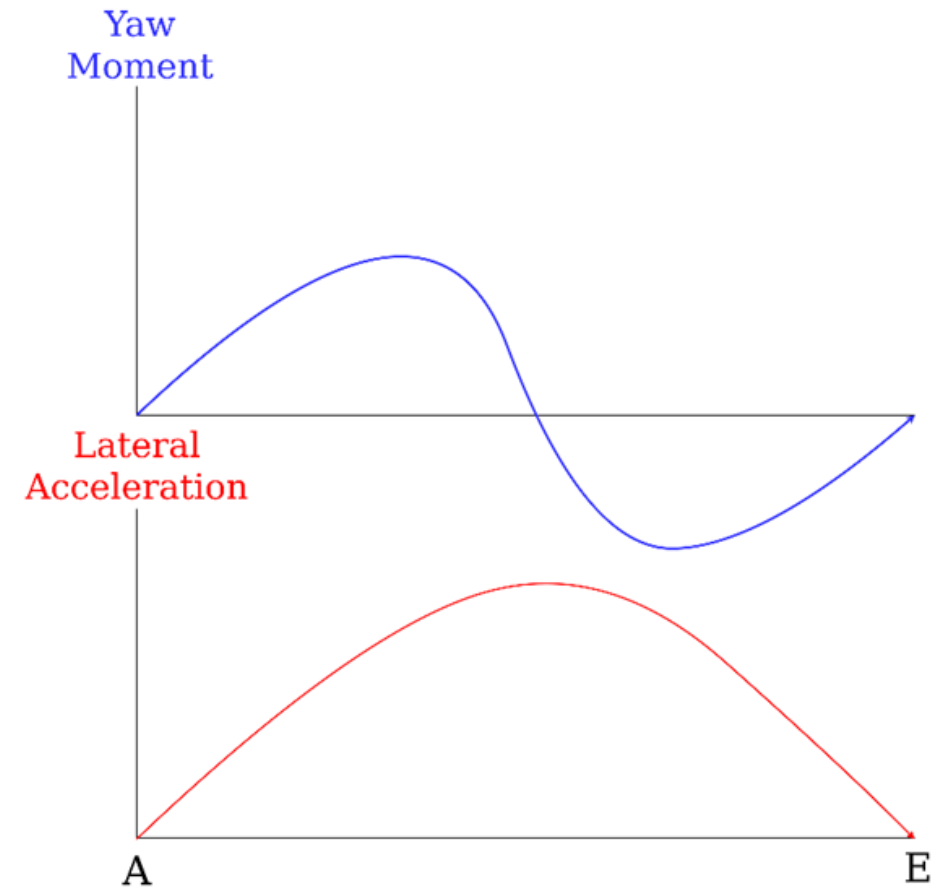
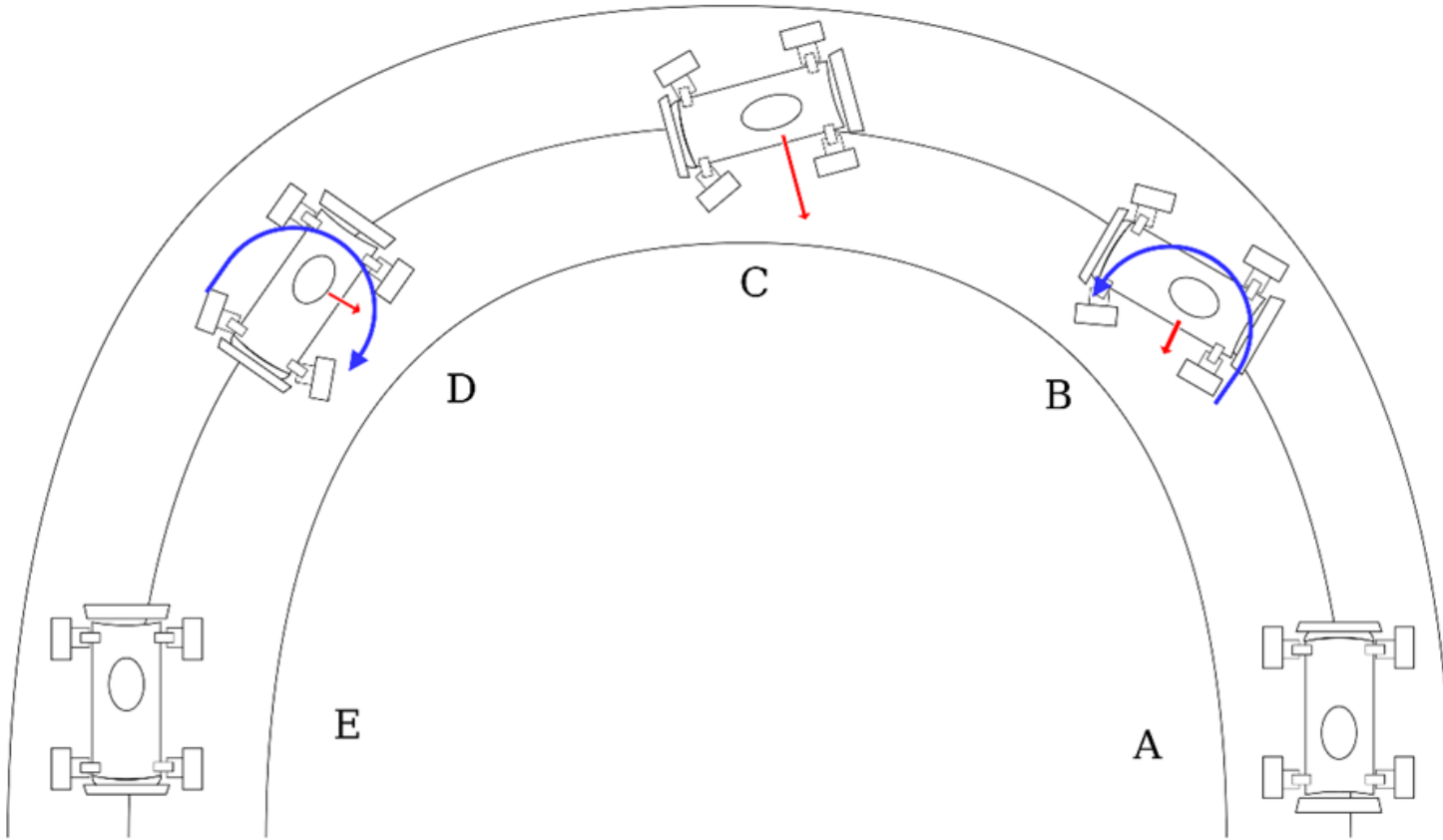


Driving Mission: Cornering

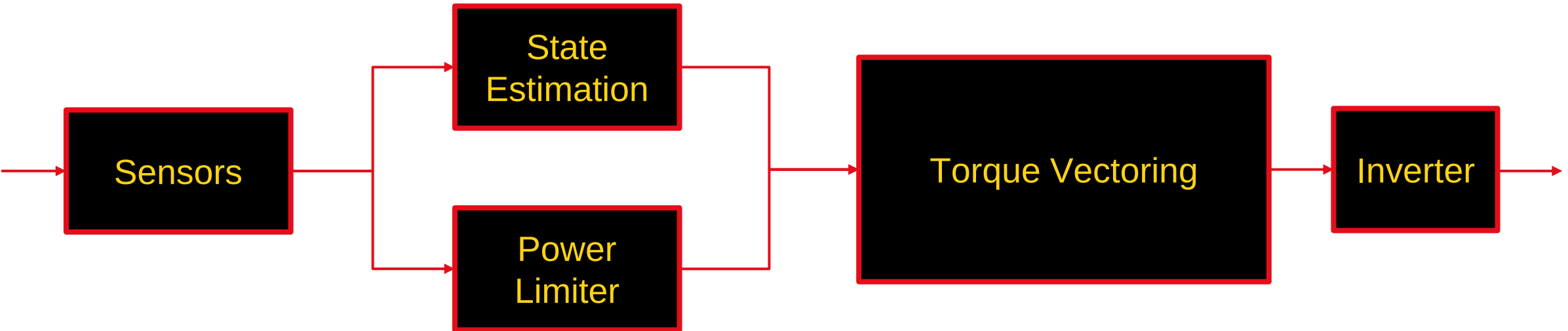
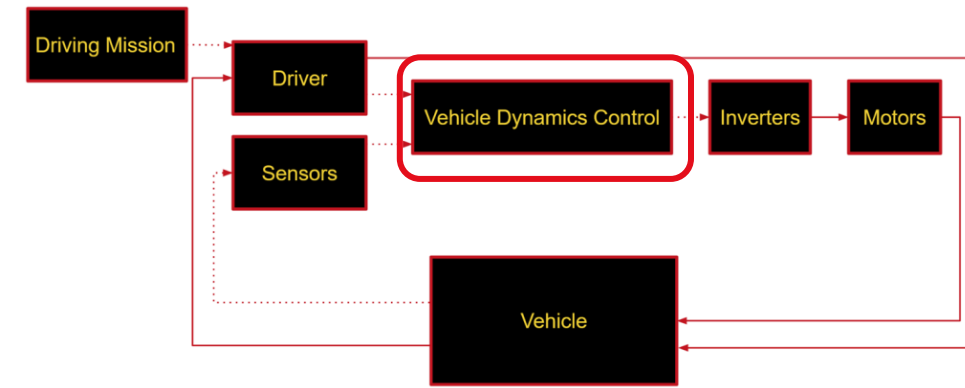
- Hairpin Example
 - 180° Turn
 - Apex (C)
 - Max lateral g
 - Steady State
 - Zero Yaw Acceleration



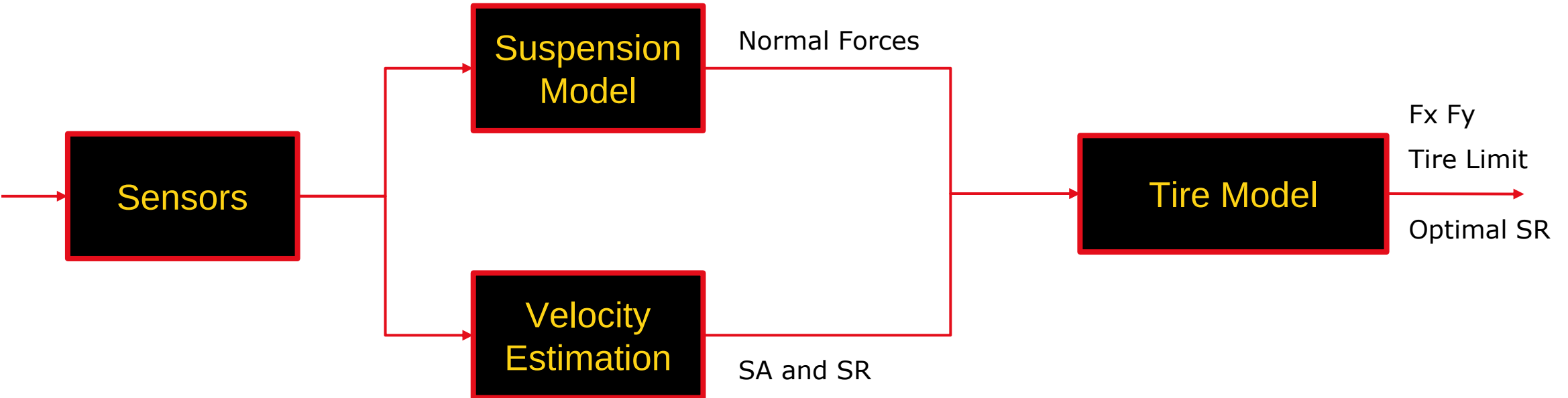
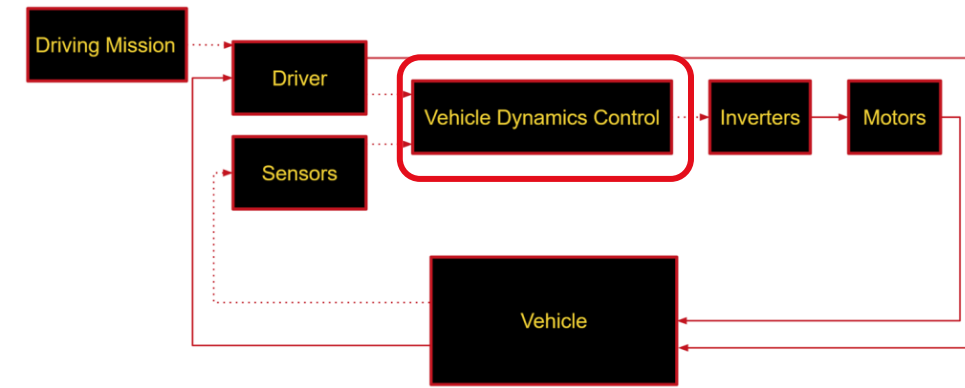
Cornering



Vehicle Dynamics Control

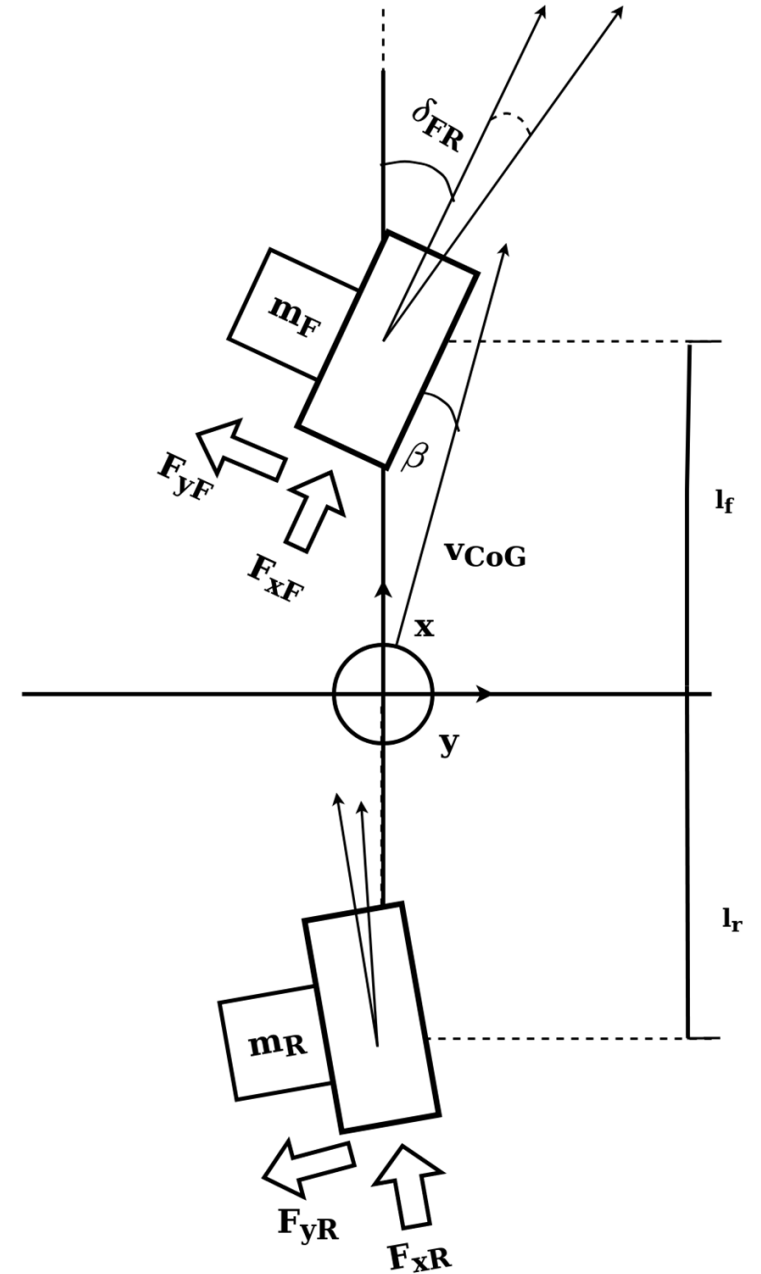


State Estimation



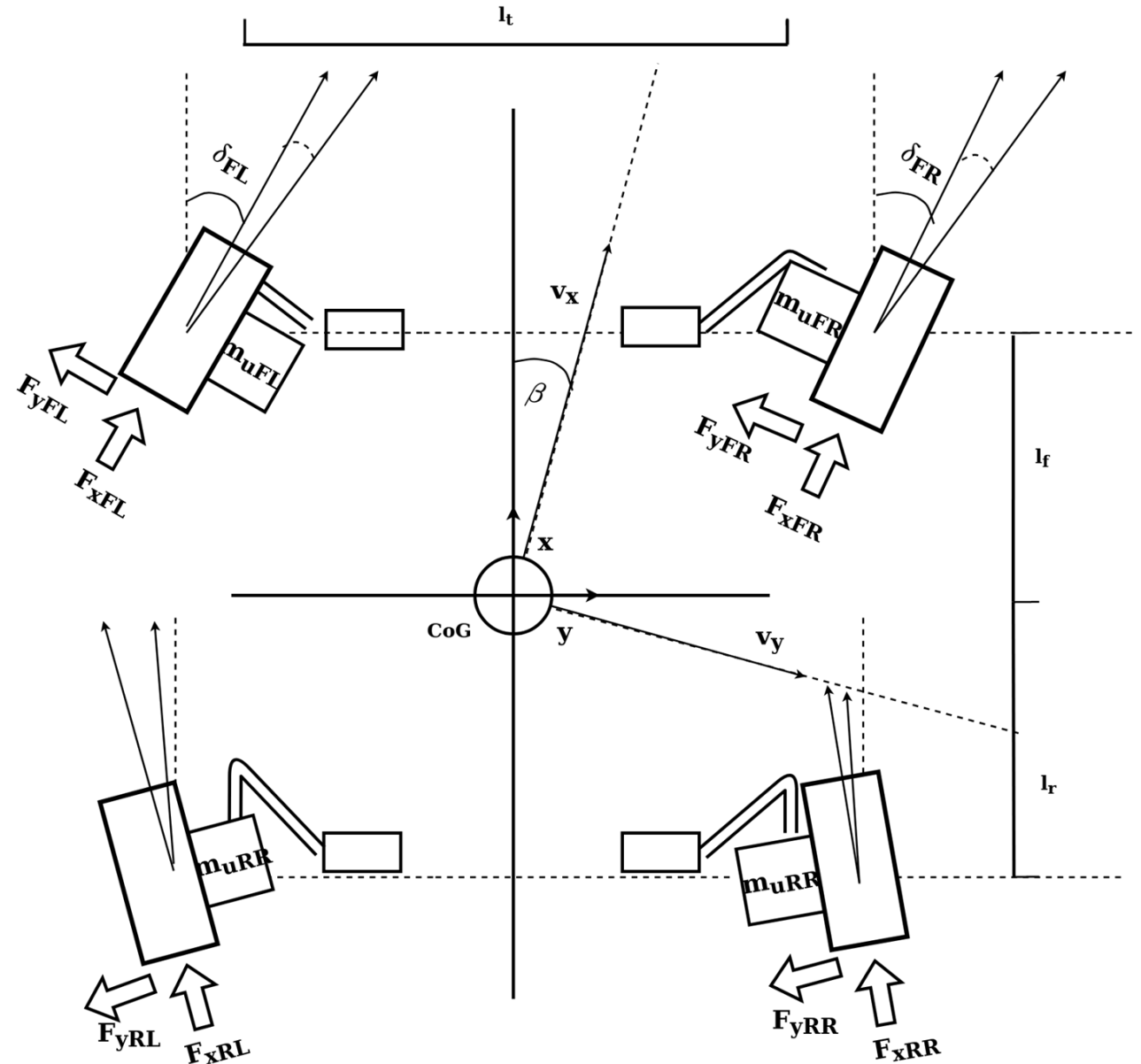
Yaw Dynamics Model

- Bicycle Model
 - Each axis modelled as single tire
 - May be used for
 - Active Yaw Moment Reference
 - Velocity Estimation



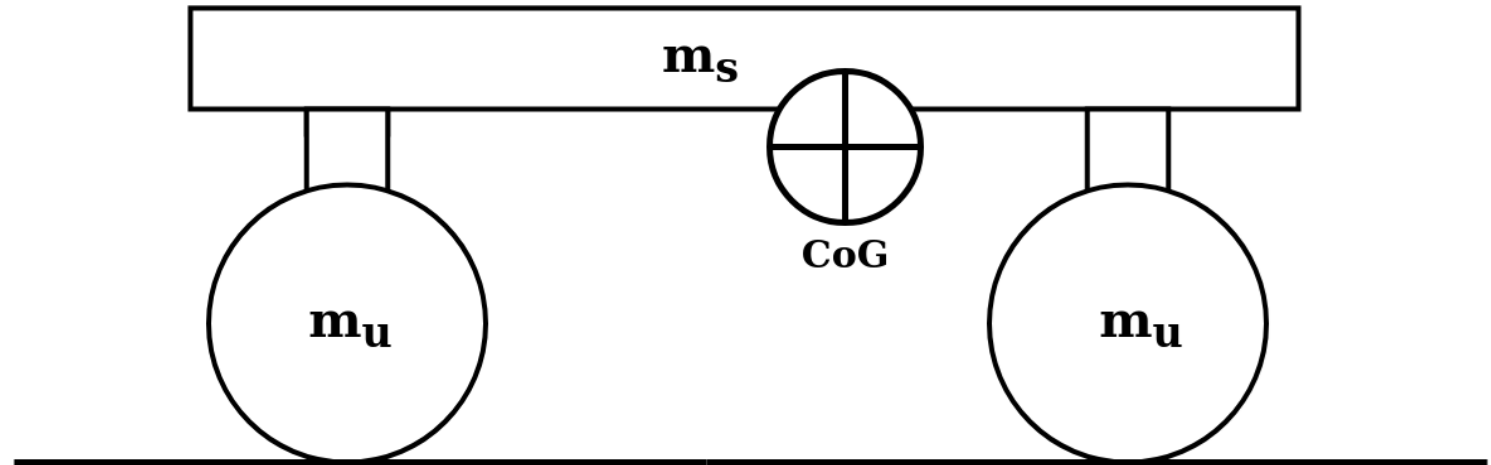
Yaw Dynamics Model

- Full Car Model
 - Necessary for
 - SR Calculation
 - SA Calculation
 - Torque Distribution



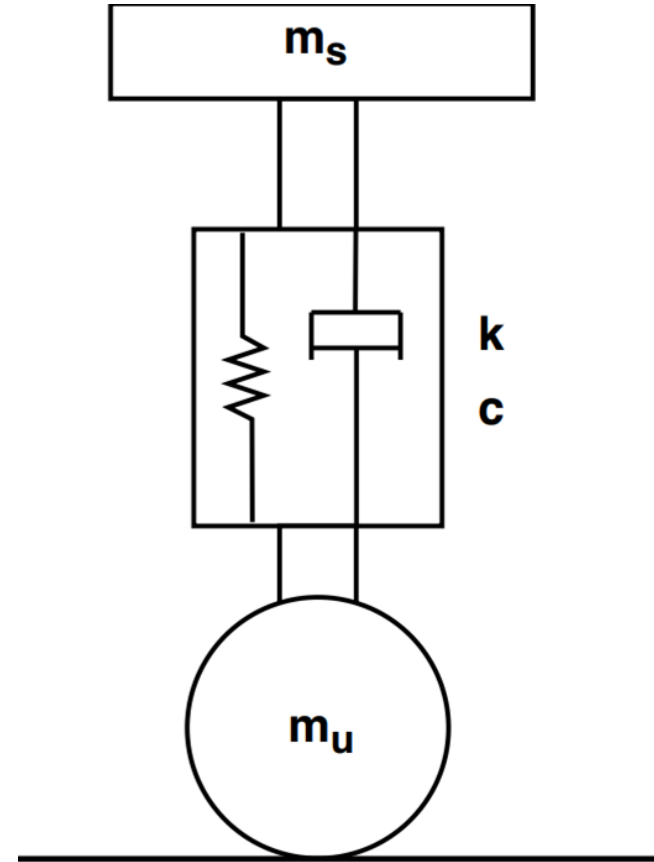
Suspension Model

- Steady State Model
 - No SDE Consideration
 - WT around CG
 - Aerodynamics



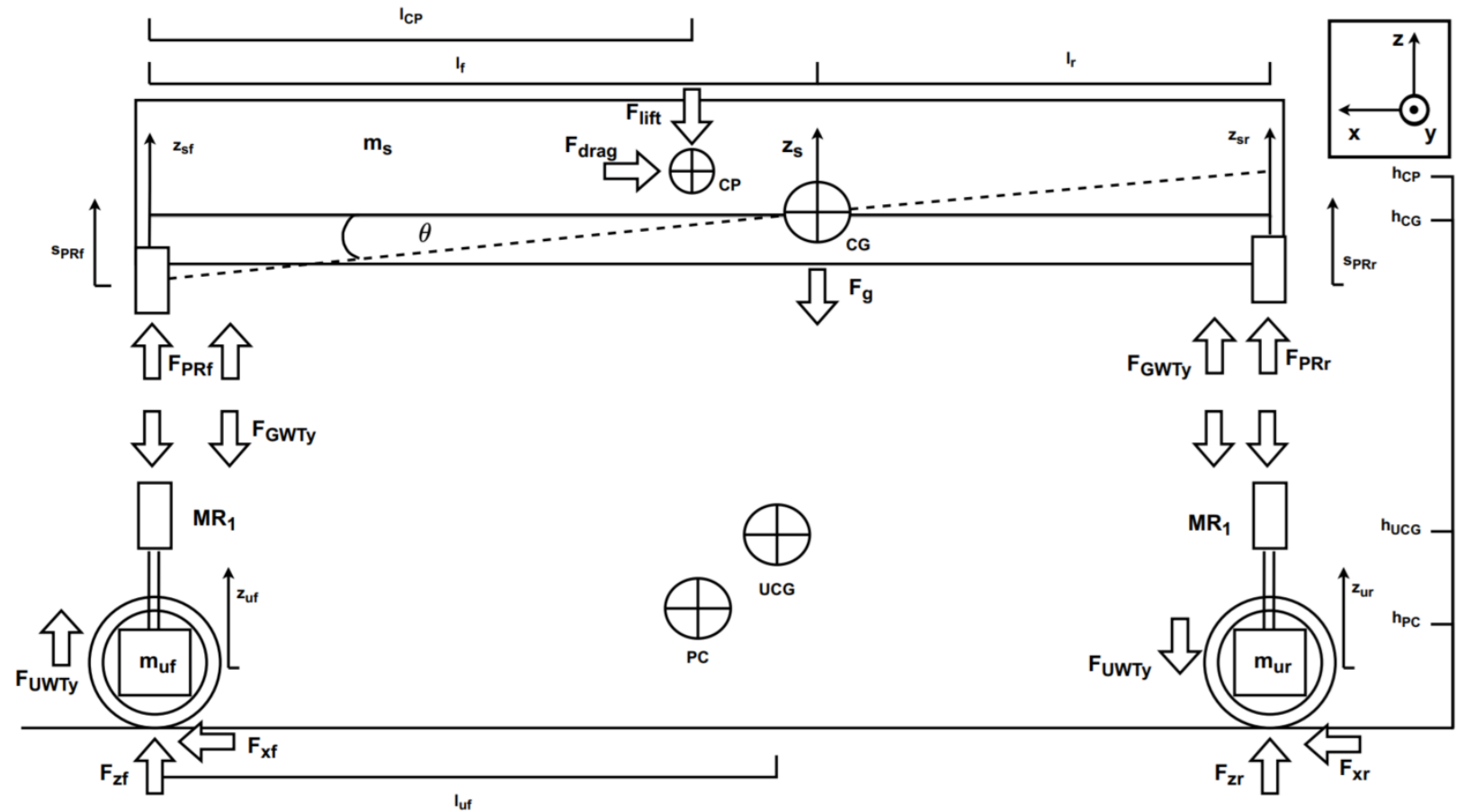
Suspension Model

- Quarter Car Model
 - SDE consideration
 - Modelling of transient effects
 - Adequate description for most mechanic suspension layouts



Suspension Model

- Full Car Model
 - Full Kinematic layout
 - Lateral Dynamics
 - Longitudinal Dynamics
 - Sophisticated Suspension Layouts

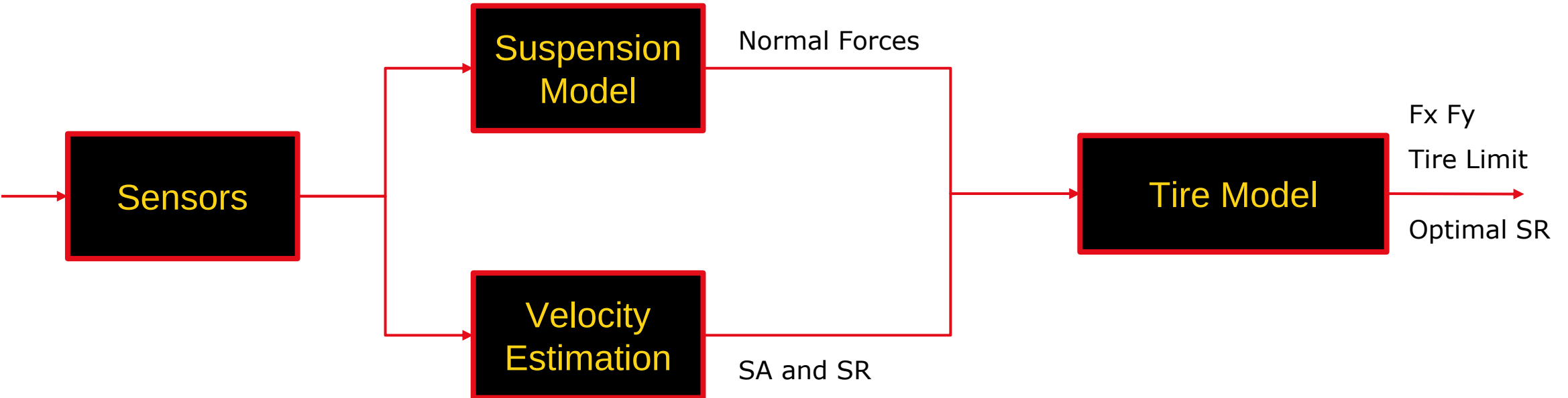
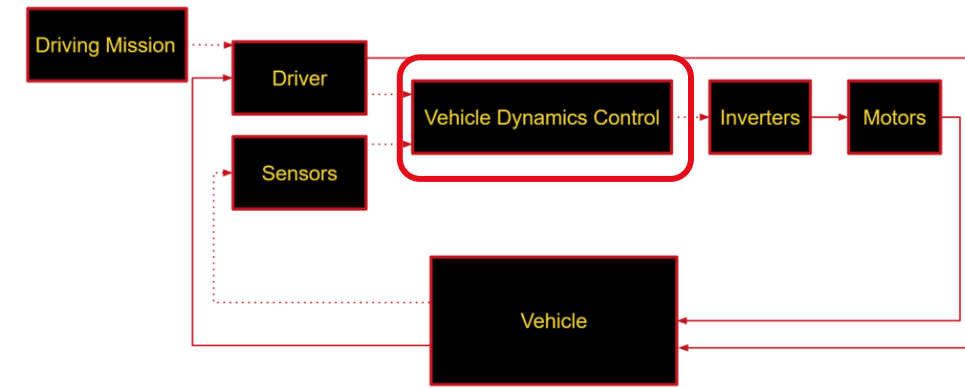


Tire Model

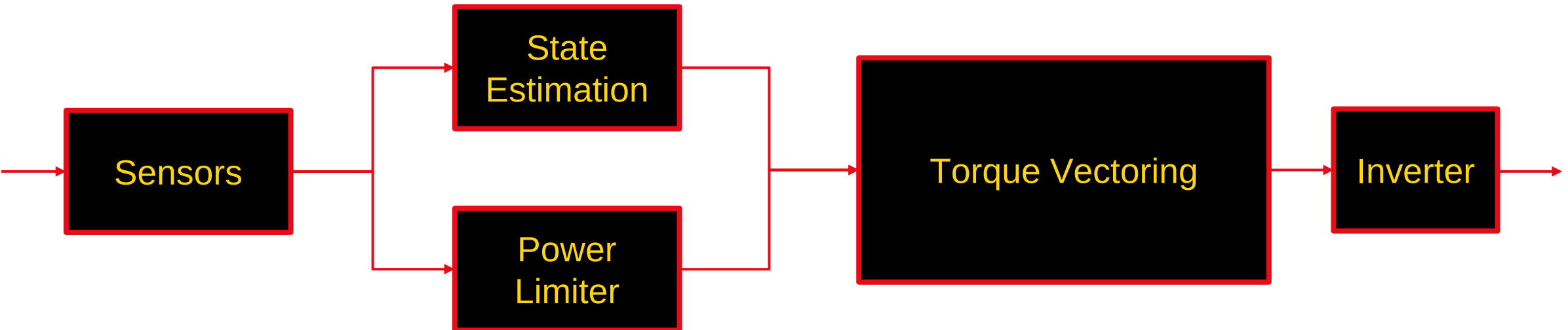
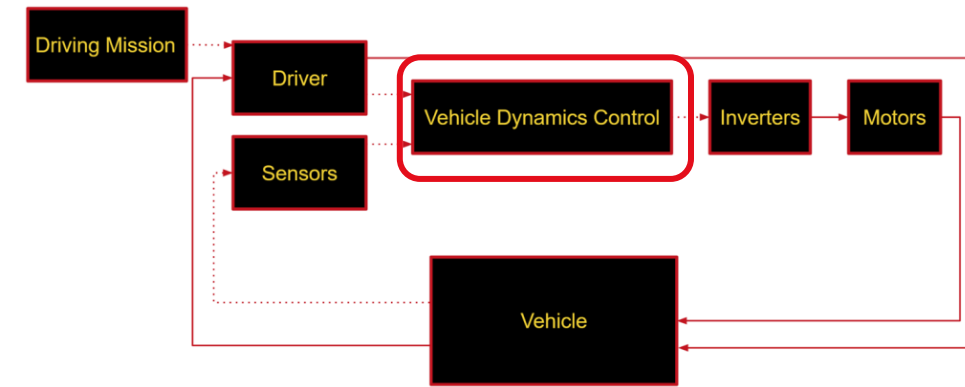
- Tire Manufacturer Data
- Independent Tire Tests
- Fit tire model to test data



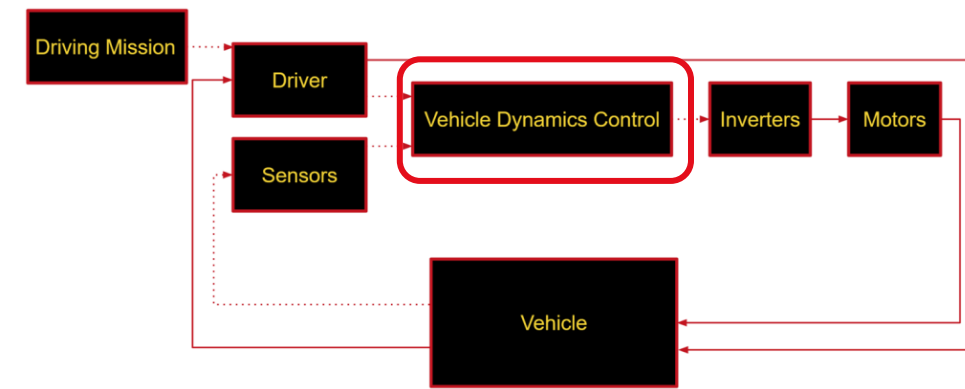
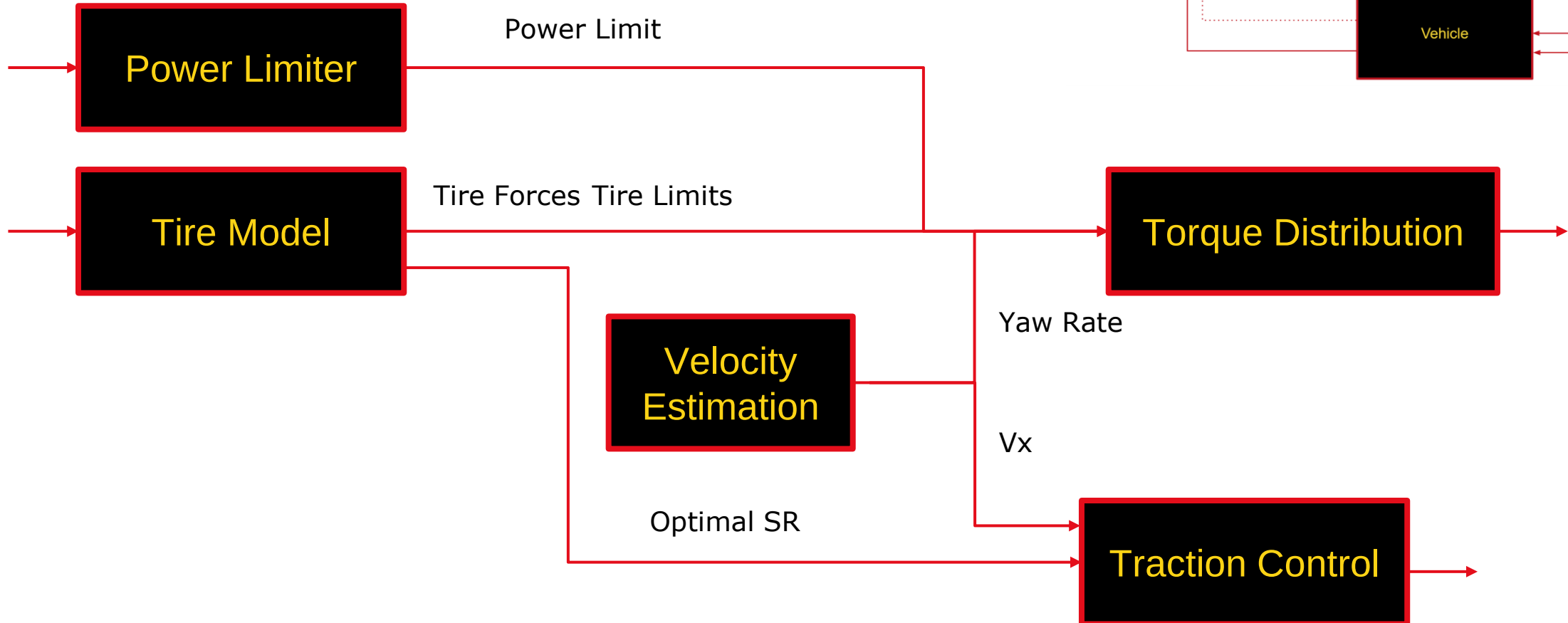
State Estimation



Vehicle Dynamics Control



Torque Vectoring



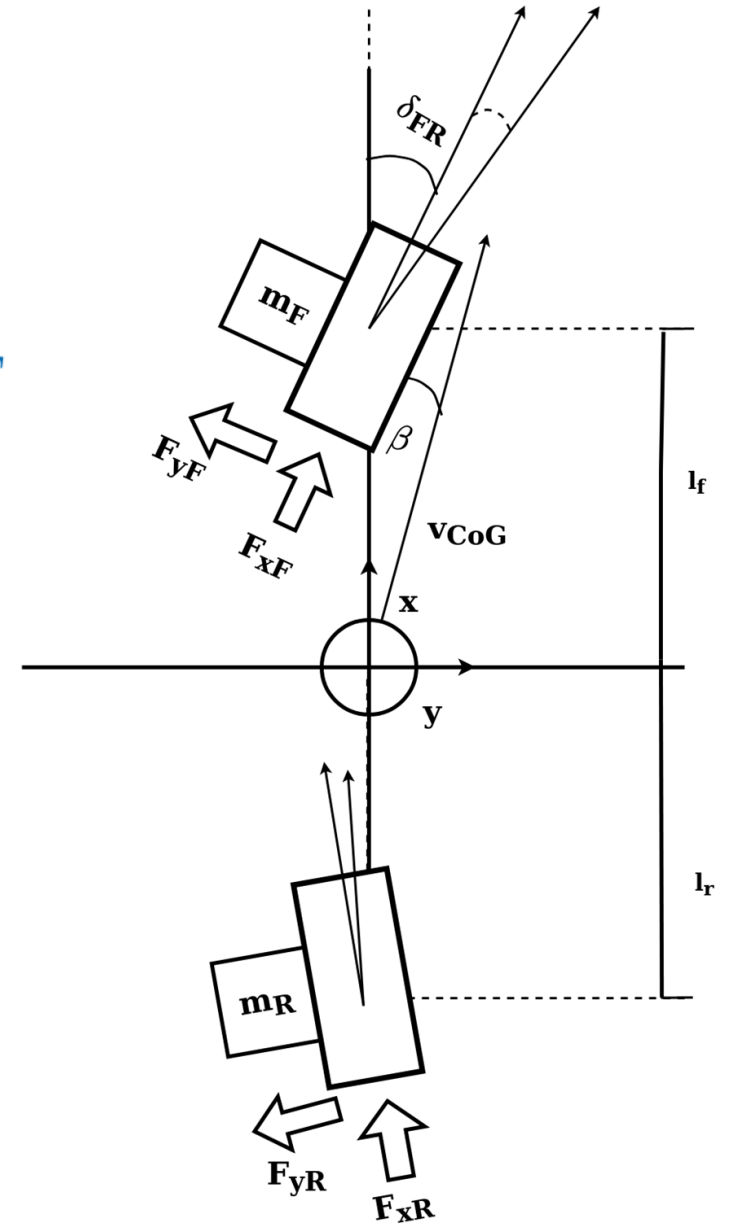
Torque Distribution: EoM

$$m_V \cdot v_H \cdot (\omega_Z + \dot{\beta}) = F_{Y,\beta} \cdot \beta + F_{Y,\omega_Z} \cdot \omega_Z + F_{Y,\delta} \cdot \delta_F$$

$$I_{ZV} \cdot \alpha_Z = M_{Z,\beta} \cdot \beta + M_{Z,\omega_Z} \cdot \omega_Z + M_{Z,\delta} \cdot \delta_F$$

$$K_{stability} = \frac{m_V \cdot M_{Z,\beta}}{L \cdot (M_{Z,\beta} \cdot F_{Y,\delta} - F_{Y,\beta} \cdot M_{Z,\delta})}$$

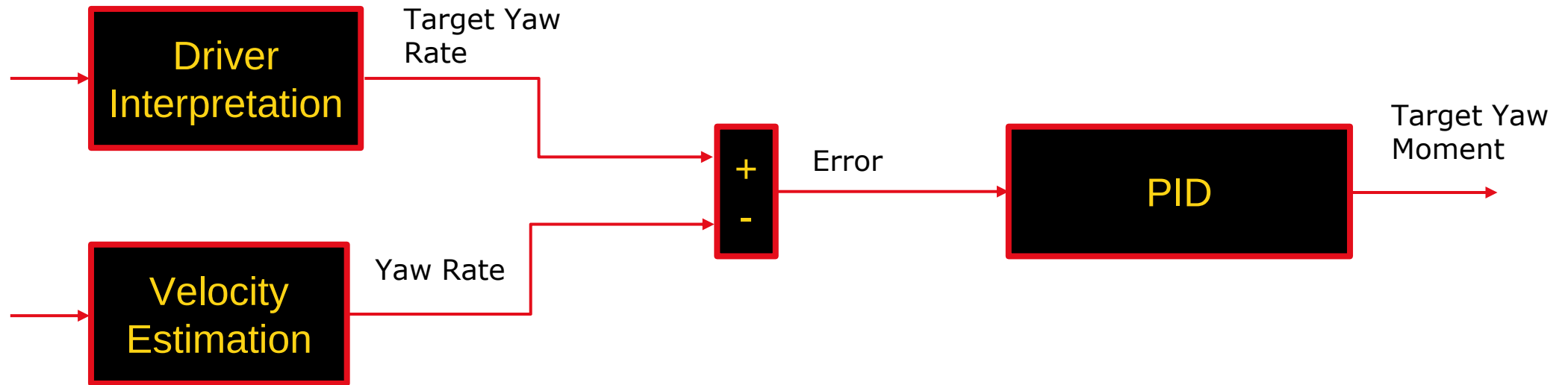
$$L = \frac{v_H \cdot (M_{Z,\beta} \cdot F_{Y,\omega_Z} - F_{Y,\beta} \cdot M_{Z,\omega_Z})}{F_{Y,\beta} \cdot M_{Z,\delta} - M_{Z,\beta} \cdot F_{Y,\delta}}$$



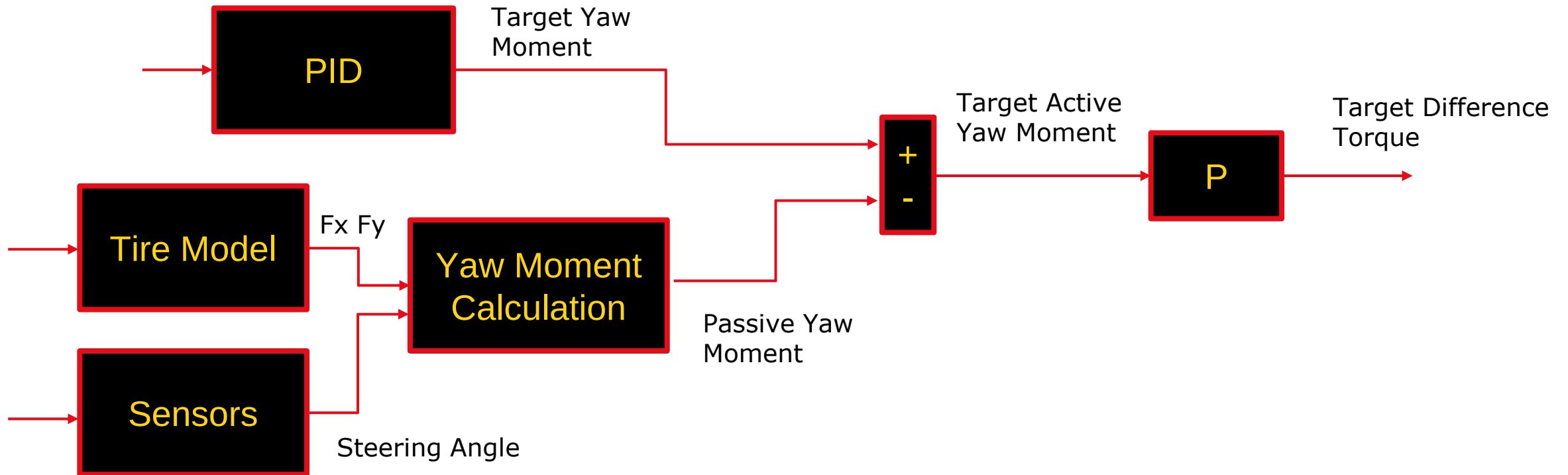
Torque Distribution: Target Yaw Rate

$$\omega_Z = \frac{\delta_F \cdot v_H}{\frac{1}{L} \cdot (1 + K_{stability} \cdot v_H^2)}$$

Torque Distribution: Target Yaw Moment



Torque Distribution: Target Difference Torque

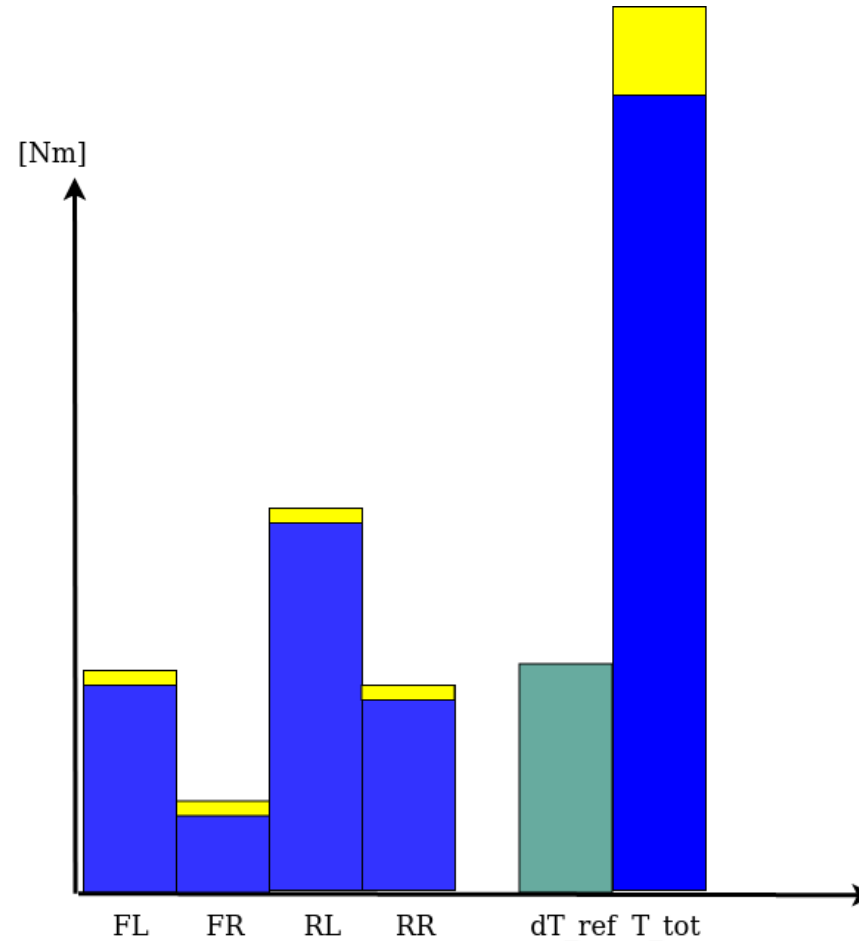


Torque Distribution: Optimization Algorithm

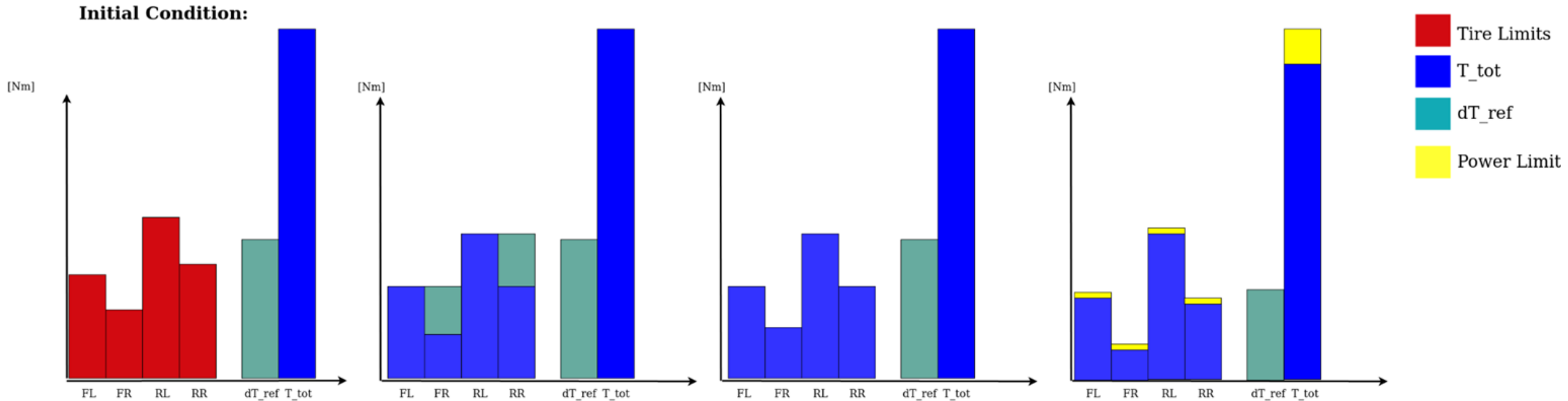
- Target
 - Reference Total Torque
 - Reference Difference Torque
- Constraints
 - Motor Limits
 - Tire Limits
 - Power Limit
- Heuristic Approaches
 - Rule Based
 - Fuzzy Logic
- Dynamic Programming
 - Infinite Horizon
- MPC

Torque Distribution: Optimization Framework

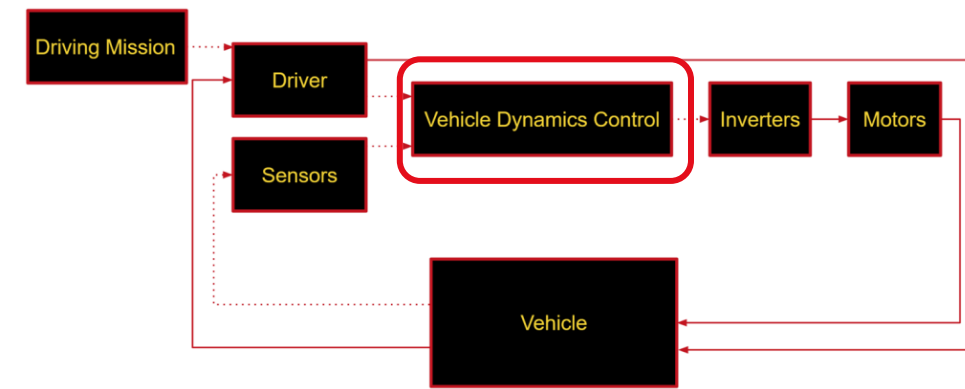
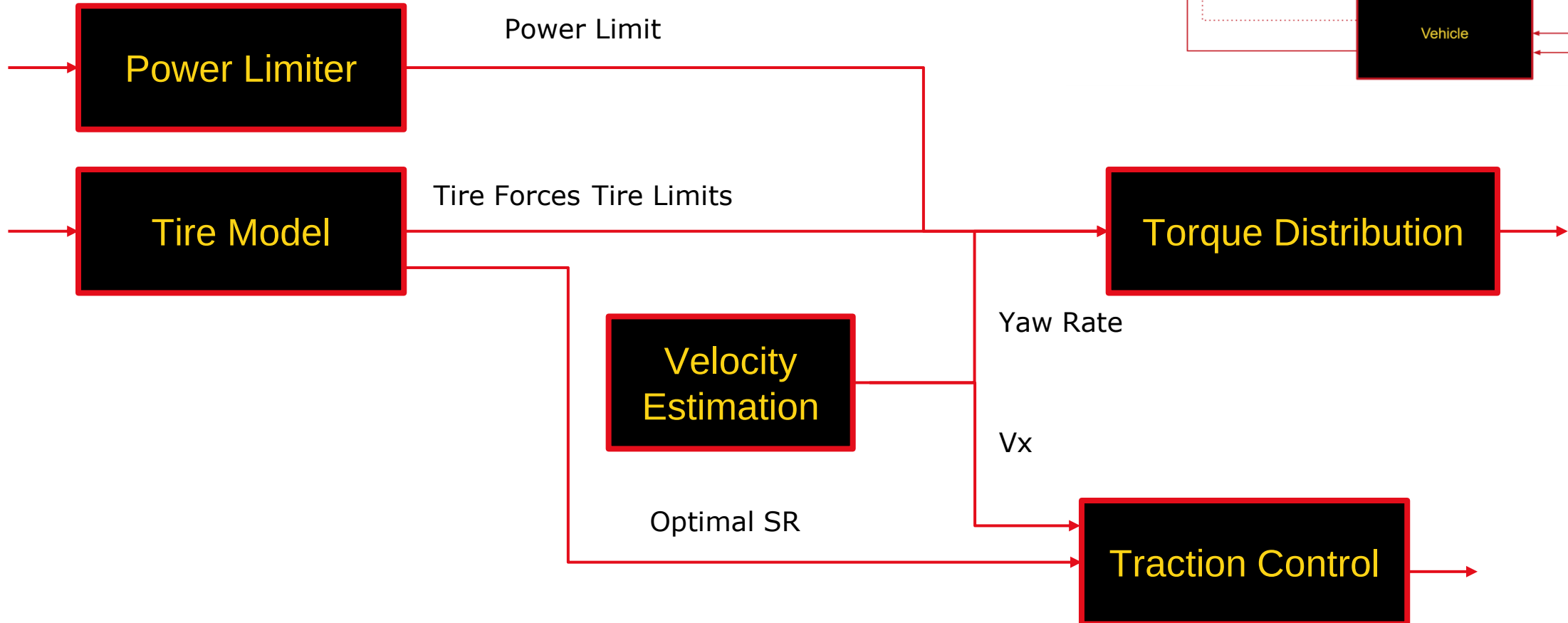
Initial Condition:



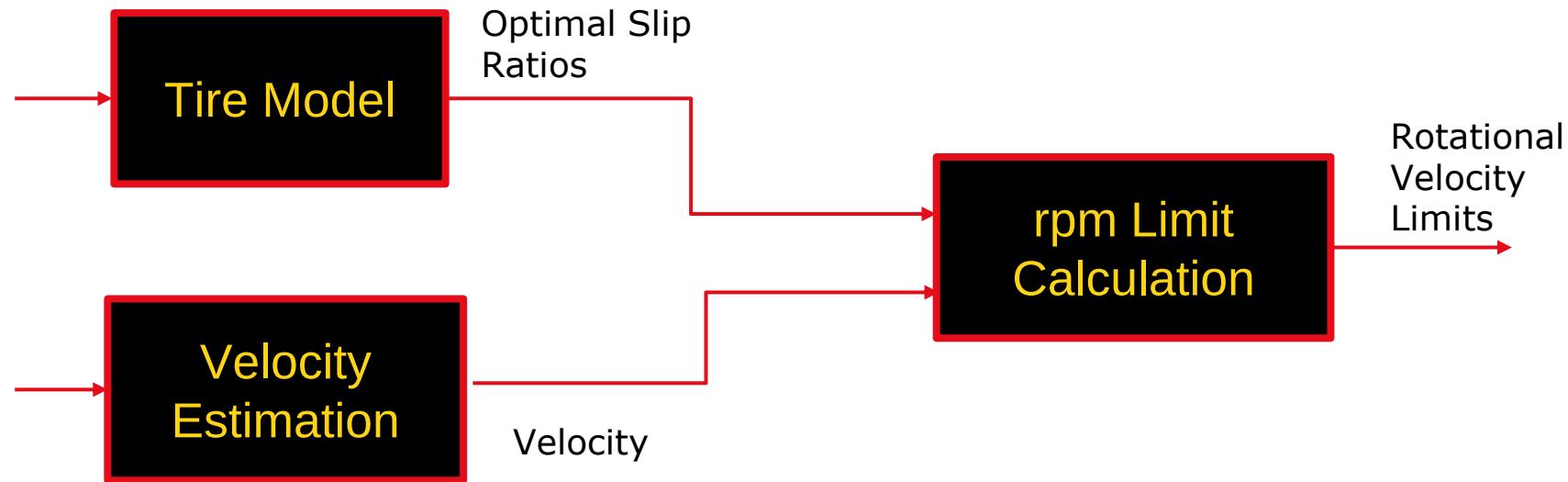
Torque Distribution: Optimization Framework



Torque Vectoring



Traction Control: Rotational Velocity Limits

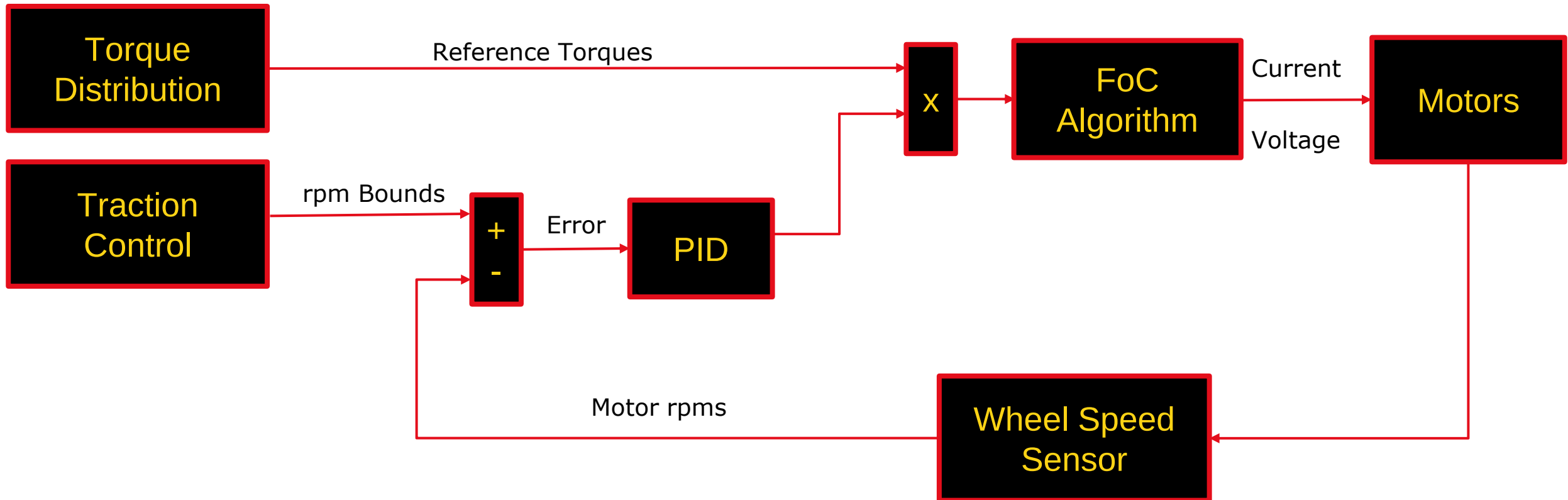
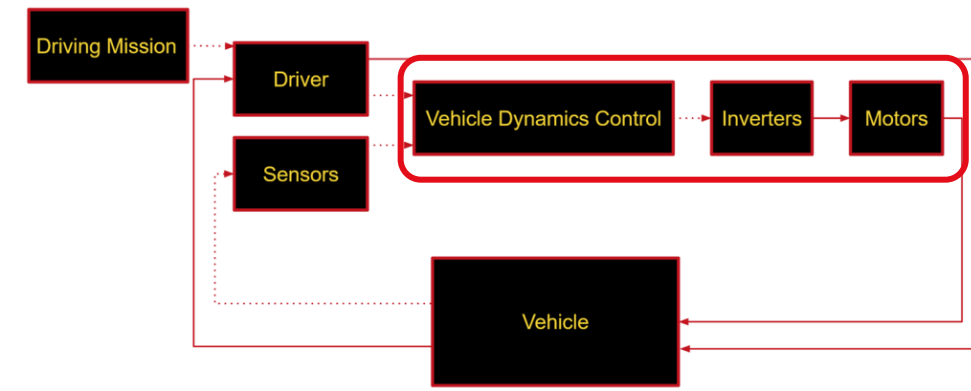


Traction Control: Rotational Velocity Limits

$$SR_{opt_{accel}} = \frac{\omega_{wheel_{max}} * r_{dyn} - v_x}{v_x}$$

$$SR_{opt_{decel}} = \frac{\omega_{wheel_{min}} * r_{dyn} - v_x}{v_x}$$

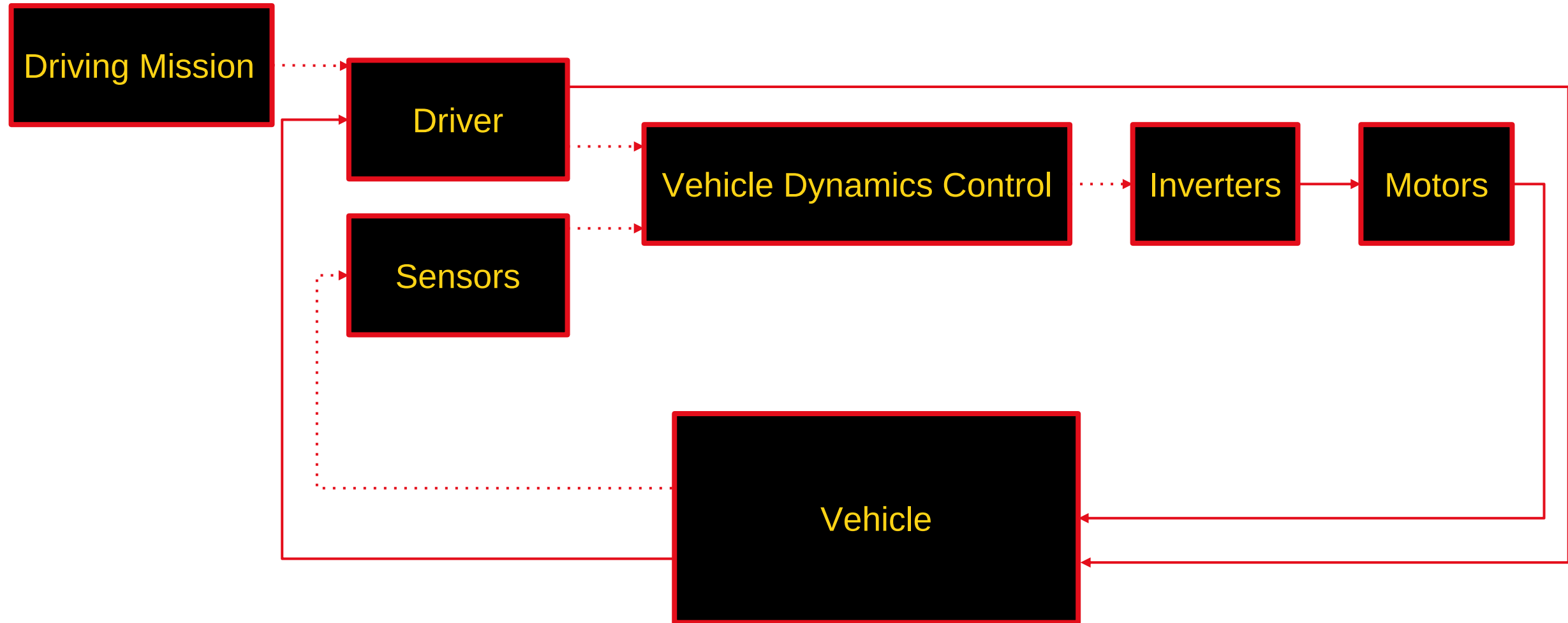
Traction Control: Inverter



Tuning

- Input Model
 - Tire Grip
 - Rotational Velocities
- Target Yaw Rate
 - PID Tuning
 - Constant Velocity Tests
 - Constant Radius Tests
- Torque Distribution
 - Optimization Algorithm
- Traction Control
 - PID Tuning

Closing Remarks



Thank you for your attention